

Doom PWAD QA Report

Run 4b6f628d-ba00-4702-9222-190a71f98a86 | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of map E1M1 using an LLM agent, designed to evaluate navigation, combat performance, and resource management. The test concluded with a 'timeout' outcome after 61 seconds, failing to reach the map exit.

Pass/fail:

map_navigation: FAIL

combat_engagement: FAIL

resource_balance: FAIL

secret_coverage: PASS

overall_verdict: PARTIAL

- The agent timed out and failed to reach the map exit or explore a significant portion of the map.
- Only 1 of 8 spawned enemies was killed, failing the 50% engagement threshold.
- Selected_skill_summary indicates health and ammo ratios of 0.0, failing the requirement of >= 0.15 health and >= 0.5 ammo.
- Secret_sector_count is 0, so secret_coverage is pass by default.

No technical problems encountered during this test run. Total LLM calls (10) equaled total actions taken (10), indicating no fallback actions were triggered due to service errors.

Test Item And Environment

The test evaluated map E1M1 running on the Freedoom1 IWAD at difficulty level 3. The map features 8 enemies (Imps, Zombiemen, and a Shotgun Guy), 1 item (Green Armor), 0 keys, and 0 secrets within a 5-sector, 2528x1056 unit area. Static complexity is moderate, characterized by a high hitscanner concentration of 62.5%.

The environment utilized ViZDoom for engine execution, the 'gemini-3.1-flash-lite' LLM model for agent decision-making, and the MCP tool interface. The run was constrained by a 500-tick cap.

Metric	Value	Metric	Value
Status	completed	Duration	61
Actions	10	LLM calls	10
Final HP	100	Kills	1
Position samples	137	Movement units	891.8
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/4b6f628d-ba00-4702-9222-190a71f98a86.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	8	8	0 hidden by skill/multiplayer flags
Items	1	1	0 hidden by skill/multiplayer flags
Secrets	0	0	Sector special 9
Geometry	35 linedefs	5 sectors	2528 x 1056 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0.

Coverage Evaluation

Coverage was insufficient for a full QA assessment. The agent only visited 5 position clusters and traversed approximately 891 units, failing to engage 7 of the 8 spawned enemies or explore beyond the immediate start area.

Objectives Covered

- **Engage visible enemy (Zombieman):**
- **Document resource balance issues:**

Objectives Omitted

- **Navigate the entire map:** navigation timeout occurred; agent was unable to leave the starting area due to pathfinding issues.
- **Engage all enemy types encountered:** 7 of 8 enemies remained unengaged due to premature termination.
- **Stress-test geometry, doors, lifts, and switches:** no interactions with map geometry occurred.

Uncovered attributes: No interactions with doors, switches, or elevators were recorded. The agent's inability to navigate complex geometry prevented testing of map-specific flow and progression mechanics.

Defects And Risks

No defects were detected during this test run. The map passed all automated checks, though the run was inconclusive due to the navigation timeout.

Patterns: Insufficient defects to identify patterns.

No defects are currently attached to this run.

Risk Areas

- **Starting position/corridor:** medium Agent encountered persistent navigation issues and 'stuck' states, suggesting potential geometry complexity that may frustrate player movement.

Activity Log Summary

1) Static analysis phase (Pre-run), 2) Map exploration phase (Ticks 0-3), 3) Combat phase (Ticks 4, 8-9), 4) Navigation troubleshooting (Ticks 5-7).

Exploration was severely hampered by navigation failures, resulting in the agent repeatedly attempting to reach the same item while stuck.

Tick	Event	Position	HP	MCP Action	Stop reason
4	kill	-90.6, -255.4	100	strafe_and_shoot	shots_complete

Limitations And Recommendations

The agent was unable to reach doors, teleporters, or any areas beyond the starting sector.

Good Quality Areas

- **Initial combat engagement:** The agent successfully identified and neutralized a Zombieman threat using the strafe_and_shoot tool, demonstrating basic functional combat capability.